Bounded critical Gaussian cost for every Boolean function at quadratic depth

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Abstract

Consider fully connected tanh networks with the critical, or mean-field, Gaussian prior: first-layer, internal-layer, and readout entry variances are $1/d$, $1/N$, and $1/N^2$, respectively. We prove that there are universal constants A, K_0 such that, at one common depth $L_d \le A(d+1)^2$, every Boolean function on d inputs has global-margin large-deviation rate at most K_0. The depth is independent of the function. The proof first bounds every eigenvalue of the initial tanh feature kernel below by $\exp[-O(d)]$. Small rank-one covariance tilts then amplify an arbitrary Boolean component at bounded total relative-entropy cost. This disproves a polynomial bound on absolute-rate sublevel cardinalities jointly in dimension and depth, even at a fixed rate threshold. It does not prove a Bayesian sample-complexity lower bound for a small-circuit target: differences between competing rates remain uncontrolled.

1 Model, meaning of the result, and proof organization

Let $\mathcal{X} = \{-1, 1\}^d$, with $d \ge 1$, and let $L \ge 2$. A width-N network has $L-1$ hidden layers and computes

$$h^0 = \bar{x} = (1, x), \qquad h^s = \tanh(W^s h^{s-1}) \quad (1 \le s < L), \qquad z_w(x) = v^T h^{L-1}.$$

There are no biases after the augmented input. Under the raw prior $\mu_{N,1}$, all weights are independent centered Gaussians, with entry variances $1/d$ in W^1, $1/N$ in internal matrices, and $1/N^2$ in v. For a Boolean target f, write

$$A_{N,f} = \{w : f(x) z_w(x) > 1/2 \text{ for every } x \in \mathcal{X}\}.$$

At fixed d, L, the limit

$$J_L(f) = \lim_{N \to \infty} -\frac{1}{N} \log \mu_{N,1}(A_{N,f})$$

exists. We reproduce its proof in Appendix A. In particular the limit is taken in width with dimension and depth fixed; no uniform interchange of these limits is asserted.

Relative entropy is denoted by $D(\nu \| \rho)$; it is infinite when ν is not absolutely continuous with respect to ρ. For a feature law ν on $[-1, 1]^{\mathcal{X}}$, define $Q(\nu) = \mathbb{E}_\nu h h^T$. Let M_Q be the law of

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coordinatewise tanh applied to a centered Gaussian with covariance Q, and set $Q_0(x,y) = (1+x\cdot y)/d$. The rate has the variational formula

$$J_L(f) = \inf\Big\{ D(\nu_1 \| \mathsf{M}_{Q_0}) + \sum_{s=2}^{L-1} D(\nu_s \| \mathsf{M}_{Q(\nu_{s-1})}) + \tfrac{1}{2} z^T Q(\nu_{L-1})^+ z : \\ f(x)z(x) \ge 1/2 \text{ for every } x \Big\}.$$

Here Q^+ is the Moore–Penrose inverse. The final term is infinite when z is outside the covariance range. All covariance matrices here are uncentered feature second moments. The constructed feature laws happen to have zero mean, so those two notions agree for the main construction.

Theorem 3.1 proves that universal finite A, K_0 exist such that, for each d, one common integer $L_d \le A(d+1)^2$ satisfies $J_{L_d}(f) \le K_0$ for every Boolean f. Thus

$$\log \#\{f : J_{L_d}(f) \le K_0\} = 2^d \log 2.$$

No polynomial in d, L can bound this logarithmic cardinality at the fixed threshold K_0. Equivalently, for every fixed d and $\eta > 0$, all functions have raw probability at least $\exp[-N(K_0+\eta)]$ once N is sufficiently large. The width threshold in this equivalent asymptotic statement is not quantified here.

We first lower-bound the initial feature spectrum. We then track a rank-one component vff^T through three stages: free variance reduction, amplification of its fraction of the covariance, and amplification of its absolute size. The total Gaussian relative entropy stays bounded, and a final readout realizes f with unit margin at bounded cost. The proof below is a conventional mathematical argument and does not claim Lean formalization. Constants are deliberately nonoptimized.

2 A quantitative first-layer spectrum bound

The proof starts with a quantitative lower bound on every Fourier eigenvalue of the first tanh feature kernel.

Lemma 2.1 (A quantitative first-layer spectrum bound)**.** *Let $d \ge 1$, $D = d+1$, $m = 2^d$, and let Q be the $m \times m$ matrix*

$$Q(x,y) = \mathbb{E}_w \tanh(w\cdot\bar{x}) \tanh(w\cdot\bar{y}), \qquad w \sim \mathcal{N}(0, I_{d+1}/d).$$

Then

$$\lambda_{\min}(Q) \ge e^{-13D}.$$

Proof. Write $q = D/d \in [1,2]$, and let h_n be the coefficient of the normalized probabilists' Hermite polynomial $\mathrm{He}_n/\sqrt{n!}$ in $z \mapsto \tanh(\sqrt{q}z)$. The coefficients vanish for even n. For every odd $n \ge 1$, the Fourier sine identity

$$\tanh(\sqrt{q}z) = \frac{1}{\sqrt{q}} \int_0^\infty \frac{\sin(tz)}{\sinh(\pi t/(2\sqrt{q}))}\, dt$$

and $\mathbb{E}[\mathrm{He}_n(Z) e^{itZ}] = (it)^n e^{-t^2/2}$ give

$$|h_n| = \frac{1}{\sqrt{qn!}} \int_0^\infty \frac{t^n e^{-t^2/2}}{\sinh(\pi t/(2\sqrt{q}))}\, dt.$$

The sine identity follows, for example, by expanding $1/\sinh(\pi t/2) = 2\sum_{j\ge 0} e^{-\pi(j+1/2)t}$ and using the logarithmic derivative of the Euler product for cosh. Fubini is valid in the displayed Hermite integral: near zero use $|\sin(tZ)| \le t|Z|$, and at infinity the reciprocal hyperbolic sine has exponential decay.

Restrict the positive integral to $[\sqrt{n}, \sqrt{n}+1]$. Using $n! \le n^n$, $1/\sinh u \ge 2e^{-u}$, and $1 \le q \le 2$ gives

$$|h_n| \ge \sqrt{2}\exp\left[-\frac{n}{2} - \left(1+\frac{\pi}{2}\right)\sqrt{n} - \frac{1+\pi}{2}\right] \ge e^{-6n}.$$

The last inequality follows from $\sqrt{n} \le n$, $n \ge 1$ and $\pi < 4$.

Mehler's identity now yields the nonnegative power series

$$Q(x,y) = \sum_{n \text{ odd}} h_n^2 \left(\frac{1+\sum_{i=1}^d x_i y_i}{D}\right)^n.$$

The Boolean characters diagonalize this translation-invariant kernel. For a character of degree r, choose the odd integer $n = r$ if r is odd and $n = r+1$ if r is even. Then $1 \le n \le D$. In the multinomial expansion of the nth power, choosing every selected coordinate exactly once, and choosing the constant coordinate once when r is even, contributes $n!/D^n$ to this character's normalized eigenvalue. Every other contribution is nonnegative. Therefore its eigenvalue for Q/m is at least

$$h_n^2 \frac{n!}{D^n} \ge e^{-12n-D} \ge e^{-13D}.$$

Here $n! \ge (n/e)^n$, and $n[1+\log(D/n)] \le D$, prove $n!/D^n \ge e^{-D}$. Multiplying by $m \ge 1$ proves the stated lower bound for Q. □

3 Bounded critical rate for all functions at quadratic depth

Covariance changes are cheaper than coherent mean shifts when the relative change is small. The following sharper capacity obstruction uses that fact. It concerns the critical variational rate, and does not compare the optimal rates of competing functions closely enough to prove a posterior sample-complexity lower bound.

Theorem 3.1 (Universal bounded rate at quadratic depth)**.** *There are universal finite constants $A, K_0 > 0$ such that, for every $d \ge 1$, there is an integer depth $L_d \le A(d+1)^2$ for which*

$$J_{L_d}(f) \le K_0 \qquad \text{for every Boolean } f : \{-1,1\}^d \to \{-1,1\}.$$

The depth L_d is the same for all functions on that cube.

We first record exact scalar formulas for a controlled feature law. Put

$$F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z), \qquad a(q) = 1 - F(q), \qquad R(q) = \frac{q[1-F(q)]^2}{F(q)} \quad (q > 0).$$

The value $R(0) = 1$ makes it continuous at zero.

Lemma 3.2 (Rank-one covariance control)**.** *Suppose a feature covariance Q has every diagonal entry equal to q and $Q \succeq v f f^T$ for $v > 0$, where f is Boolean. For any $\epsilon \ge 0$, let the next feature law be coordinatewise tanh of a centered Gaussian with covariance $Q + \epsilon v f f^T$. Its entropy relative*

to the ordinary next-layer law M_Q is at most $\epsilon^2/4$. Its second moment Q' has constant diagonal q' and obeys $Q' \succeq v' f f^T$, where

$$u = q + \epsilon v, \qquad q' = F(u), \qquad v' = (1+\epsilon)v[1 - F(u)]^2. \tag{1}$$

Consequently, for $w = v/q$ and $w' = v'/q'$,

$$w' = w\frac{1+\epsilon}{1+\epsilon w}R(u).$$

All these scalar quantities depend only on q, v, ϵ, not on the other entries of Q or on the choice of f.

Proof. The Loewner bound implies $f \in \mathrm{range}(Q)$ and $v f^T Q^+ f \le 1$. The two Gaussian laws have the same support, and the rank-one determinant identity gives their relative entropy as

$$\frac{1}{2}\{t - \log(1+t)\}, \qquad t = \epsilon v f^T Q^+ f \le \epsilon.$$

This is at most $t^2/4 \le \epsilon^2/4$. Applying coordinatewise tanh can only decrease relative entropy. Write the changed Gaussian as $\sqrt{(1+\epsilon)v}\, fZ + G$, where Z is standard Gaussian and G is an independent centered Gaussian of covariance $Q - v f f^T$. Its conditional feature mean given Z is $f\psi(Z)$, since the variances of the coordinates of G all equal $q - v$. Thus $Q' \succeq \mathbb{E}\psi(Z)^2 f f^T$. Gaussian integration by parts and projection onto the first Hermite polynomial give

$$\mathbb{E}\psi(Z)^2 \ge (\mathbb{E}Z\psi(Z))^2 = (1+\epsilon)v\,[\mathbb{E}\,\mathrm{sech}^2(\sqrt{u}Z)]^2.$$

This is the claimed v', and the diagonal formula and ratio identity follow directly. In particular, these formulas automatically retain $0 < v' \le q'$, since they arise from a valid covariance lower bound. □

Proof of Theorem 3.1. Write $D = d + 1$ and $m = 2^d$. Begin with the unperturbed first feature law. Its covariance Q has constant diagonal $q_0 = F(D/d) \in [F(1), F(2)]$, and Lemma 2.1 implies

$$Q \succeq e^{-20D} I \succeq v_0 f f^T, \qquad v_0 = e^{-20D}/m.$$

This first layer costs zero entropy. We will use Lemma 3.2 at each subsequent layer, tracking its guaranteed scalar component v, not the largest possible rank-one component of the covariance.

We collect universal scalar estimates. The function F is increasing, $0 < F(q) < q$ for $q > 0$, and Taylor expansion gives

$$F(q) = q - 2q^2 + O(q^3), \qquad R(q) = 1 + O(q^2).$$

There consequently exist universal $u_0 > 0$ and $C \ge 1$ such that, for $0 < u \le u_0$,

$$F(u) \le u - u^2, \qquad R(u) \ge 1 - Cu^2. \tag{2}$$

We may and do take $u_0 \le 1/4$. A useful global lower bound is

$$F(q) \ge \frac{q}{1+3q}. \tag{3}$$

Indeed $\tanh^2 t \ge t^2/(1+t^2)$ follows from $|\sinh t| \ge |t|$. Cauchy–Schwarz applied to $V = qZ^2$ gives $\mathbb{E}[V/(1+V)] \ge (\mathbb{E}V)^2/\mathbb{E}[V(1+V)] = q/(1+3q)$.

Choose a sufficiently small universal $v_* > 0$ satisfying

$$v_* \le \min\{u_0/4, 1/64, 1/(100C)\}.$$

Then choose a positive universal c so small that

$$c \le \min\{v_*/4, u_0/3, 1/3, 1/(144C)\}.$$

Put $\delta = c/D$.

Free cooling. Apply the unmodified next-layer law, that is, $\epsilon = 0$, until $q \le \delta$ for the first time. This costs no entropy. There is a universal $c_F > 0$ with $q - F(q) \ge c_F q^2$ for $0 < q \le F(2)$: the ratio is positive and continuous, and tends to two at zero. Hence the number of cooling layers is at most a universal constant times $1/\delta$. The final cooled variance q_c lies in $[\delta/2, \delta]$, by monotonicity and (3).

During cooling the ratio $w = v/q$ is multiplied by $R(q)$ at each step. This loses at most a fixed positive factor, uniformly in the number of cooling layers. To see this directly, on $[0, F(2)]$ one has $|\log R(q)| \le C_R q^2$ for a universal C_R, by continuity, positivity and the expansion at zero. Also $\sum q_s^2 \le c_F^{-1} \sum (q_s - q_{s+1}) \le F(2)/c_F$. Thus the cooled ratio satisfies

$$w_c \ge c_0 v_0, \qquad \log(1/w_c) \le A_0 D$$

for universal $c_0 > 0, A_0 < \infty$.

Amplifying the relative component. Now set $\epsilon = \delta$ at each layer until $w \ge 1/2$. Throughout this phase $q \le 2\delta$. Indeed $u = q + \delta v \le (1+\delta)q$, and the increasing function $u - u^2$ on the relevant interval gives

$$q' \le 2\delta(1+\delta) - [2\delta(1+\delta)]^2 \le 2\delta.$$

While $w \le 1/2$, the exact ratio identity and $u \le 3\delta$ imply

$$\log(w'/w) \ge \frac{\delta}{4} - 18C\delta^2 \ge \frac{\delta}{8}.$$

Here we used $\log(1 - Cu^2) \ge -2Cu^2$, valid by the chosen constants. The number τ of these layers is therefore at most $8A_0 D/\delta + 1 = O(D^2)$. Their total entropy is at most

$$\tau\delta^2/4 \le 2A_0 D\delta + \delta^2/4 = O(1).$$

Furthermore, since $u \ge q$, monotonicity and (3) imply $1/q' \le 1/q + 3$ at every layer. Thus, at the end of this phase,

$$q \ge \frac{1}{2/\delta + 3\tau} \ge \frac{c_1}{D^2}, \qquad v \ge \frac{q}{2} \ge \frac{c_1}{2D^2}$$

for some universal $c_1 > 0$. At the same time $v \le q \le 2\delta < v_*$.

Growing the absolute component. Set $\epsilon = 16v$ until $v \ge v_*$. The invariant $w \ge 1/3$, or equivalently $q \le 3v$, is preserved. To check this, while $v < v_*$ one has $u = q + 16v^2 \le 4v$. If $w \ge 1/2$, the ratio identity gives $w' \ge \frac{1}{2}(1 - 16Cv^2) \ge 1/3$. If $1/3 \le w \le 1/2$, it gives

$$\log(w'/w) \ge 4v - 32Cv^2 \ge 0.$$

The initial ratio is at least $1/2$, so induction proves the invariant. Using $F(u) \le u$ also gives

$$v' \ge v(1+16v)(1-2u) \ge v(1+16v)(1-8v) \ge v + 4v^2, \qquad v' \le v + 16v^2.$$

Thus $1/v - 1/v' \geq 2$ until the stopping time, which is $O(D^2)$. The final value is between v_* and $2v_*$, and the total entropy of this phase telescopes:

$$\sum \frac{(16v)^2}{4} = 64 \sum v^2 \leq 16 \sum (v' - v) \leq 32 v_*.$$

Finally choose output $z = f$. Its signed margin is one, and $Q \succeq v_* f f^T$ bounds its Gaussian readout cost by $1/(2v_*)$. All entropy costs plus this readout cost are bounded by a universal constant K_0. The number of hidden layers, and therefore the depth, is at most a universal constant times D^2. Crucially, q_0, v_0, every subsequent scalar recursion, every control, and every stopping rule depend only on d. Although the full covariance matrices depend on f, the resulting depth L_d is common to all Boolean functions. This proves the theorem. □

This construction sharpens an absolute-rate capacity obstruction. It still does not show that the optimal rate of a complex competitor is below, or sufficiently close to, the optimal rate of a specified simple target. That comparison remains essential for a posterior lower bound.

4 What the capacity obstruction does not establish

The result is about absolute rates. Under the prior conditioned on global output margin and on a fixed noiseless training sample S, the exponential posterior rate of a consistent function is

$$J_L(f) - \min_{g \text{ consistent with } S} J_L(g).$$

This follows from the finite sum of global-margin function probabilities, as detailed in the appendix. A universal upper bound on all the $J_L(f)$ does not control these differences. A positive rate gap, however small, is multiplied by N and can suppress a competitor in the width limit. Nor does the theorem assert that all Boolean functions have small circuits. Consequently it is neither a simple-target learning lower bound nor a claim that the conditioned prior is uniform over Boolean functions.

The restriction to the critical Gaussian scaling also matters. This construction exploits the relative-entropy cost of changing a whole hidden feature law at the same speed N as the output event. It does not identify the corresponding rate or posterior at exponents $\gamma > 1$ or $\gamma < 1$.

A An exact variational principle at the critical exponent

For completeness we reproduce the finite-domain variational foundation. Here $\phi = \tanh$. For arbitrary fixed depth, the critical prior admits a direct finite-domain large-deviation description. This result requires neither a propagation-of-chaos assumption nor differentiability of the activation. It identifies the functional used in the critical-rate capacity constructions with the actual Gaussian probability rate. Covariance-process large deviations for Gaussian networks have been studied in a more general functional setting by Andreis, Bassetti, and Hirsch [1]. No novelty claim is made for the general large-deviation framework; the finite-domain proof below fixes the precise normalizations and strict-margin convention used here.

Fix $\gamma = 1$, enumerate $\mathcal{X} = \{x^1, \ldots, x^m\}$, where $m = 2^d$, and put $r = L - 1$. All limits in this section hold with d and L fixed. Write $E = [-1, 1]^m$ and let $\mathcal{P}(E)$ carry its weak topology. For $\nu \in \mathcal{P}(E)$ define

$$K(\nu) = \int_E h h^{\mathsf{T}} \, d\nu(h), \qquad Q_0(a, b) = \frac{\bar{x}^a \cdot \bar{x}^b}{d}.$$

For any positive semidefinite $m \times m$ matrix Q, let

$$\mathsf{T}_Q(u) = \big(\phi((Q^{1/2}u)_a)\big)_{a=1}^m, \qquad \mathsf{M}_Q = (\mathsf{T}_Q)_{\#}\mathcal{N}(0, I_m).$$

Here $T_{\#}\rho$ is the distribution of $T(U)$ when U has distribution ρ, and $Q^{1/2}$ is the positive semidefinite square root. Relative entropy is

$$D(\nu\|\rho) = \begin{cases} \displaystyle\int \log \frac{d\nu}{d\rho}\, d\nu, & \nu \ll \rho, \\ +\infty, & \text{otherwise.} \end{cases}$$

Finally define the Gaussian energy

$$\mathcal{G}_Q(z) = \begin{cases} \frac{1}{2} z^{\mathsf{T}} Q^{+} z, & z \in \operatorname{range}(Q), \\ +\infty, & z \notin \operatorname{range}(Q), \end{cases}$$

where Q^+ is the Moore–Penrose inverse.

Theorem A.1 (Critical finite-domain large deviations)**.** *Let $Z_N = (z_w(x^a))_{a=1}^m$ under the unconditioned prior $\mu_{N,1}$. The sequence Z_N satisfies a large-deviation principle with speed N and good rate function*

$$I(z) = \inf_{\nu_1,\dots,\nu_r \in \mathcal{P}(E)} \left\{ D(\nu_1\|\mathsf{M}_{Q_0}) + \sum_{s=2}^{r} D\big(\nu_s\|\mathsf{M}_{K(\nu_{s-1})}\big) + \mathcal{G}_{K(\nu_r)}(z) \right\}. \tag{4}$$

In particular, for every open $O \subseteq \mathbb{R}^m$ and closed $F \subseteq \mathbb{R}^m$,

$$-\inf_{z\in O} I(z) \le \liminf_{N\to\infty} \frac{1}{N} \log \mathbb{P}(Z_N \in O), \qquad \limsup_{N\to\infty} \frac{1}{N} \log \mathbb{P}(Z_N \in F) \le -\inf_{z\in F} I(z).$$

The infimum in (4) is attained whenever it is finite. No covariance matrix in the statement is required to be nonsingular.

Proof. We give an exact representation of the finite-width law and then apply Sanov's theorem and continuous contraction [2].

Let $\Gamma_m = \mathcal{N}(0, I_m)$. Independently for each $s = 1, \dots, r$, let $U_{s,1}, \dots, U_{s,N}$ have law Γ_m, and set

$$\eta_{s,N} = \frac{1}{N} \sum_{i=1}^{N} \delta_{U_{s,i}}.$$

Also let $V_N = G/\sqrt{N}$, where $G \sim \Gamma_m$ is independent of these empirical measures. Recursively define

$$\nu_{1,N} = (\mathsf{T}_{Q_0})_{\#}\eta_{1,N}, \qquad \nu_{s,N} = (\mathsf{T}_{K(\nu_{s-1,N})})_{\#}\eta_{s,N} \quad (2 \le s \le r).$$

These are exactly the joint laws of the network's empirical hidden-activation vectors on $\mathcal{X}$. Indeed, first-layer preactivation vectors are independent Gaussians with covariance Q_0. Conditional on the preceding layer, each subsequent row of weights gives an independent Gaussian vector with covariance

$$\frac{1}{N} \sum_{i=1}^{N} \big(h_i^{s-1}(x^a) h_i^{s-1}(x^b)\big)_{a,b} = K(\nu_{s-1,N}).$$

The conditional distribution depends on the previous layer only through its empirical measure. Finally, readout weights have variance N^{-2}, so

$$Z_N \overset{\text{law}}{=} K(\nu_{r,N})^{1/2} V_N. \tag{5}$$

This representation is exact for every N.

Sanov's theorem on $\mathcal{P}(\mathbb{R}^m)$ gives the good rate $D(\eta\|\Gamma_m)$ for each $\eta_{s,N}$. The elementary Gaussian large-deviation principle gives rate $\|v\|_2^2/2$ for V_N. Independence and the finite-product principle therefore give the good rate

$$\sum_{s=1}^{r} D(\eta_s\|\Gamma_m) + \frac{1}{2}\|v\|_2^2$$

for $(\eta_{1,N}, \dots, \eta_{r,N}, V_N)$.

The recursive map from these variables to $(\nu_1, \dots, \nu_r, z)$ is continuous. To verify the only potentially delicate point, suppose $Q_j \to Q$ and $\eta_j \Rightarrow \eta$ weakly. For every continuous $b : E \to \mathbb{R}$, the functions $b \circ \mathsf{T}_{Q_j}$ converge uniformly on each compact subset of $\mathbb{R}^m$ to $b \circ \mathsf{T}_Q$, since matrix square roots are continuous and ϕ is continuous. Their absolute values are uniformly bounded. Tightness of a weakly convergent sequence of probability measures then gives

$$\int b \circ \mathsf{T}_{Q_j}\, d\eta_j \longrightarrow \int b \circ \mathsf{T}_Q\, d\eta.$$

Also $\nu \mapsto K(\nu)$ is continuous because every coordinate $h_a h_b$ is continuous and bounded on E. This proves continuity of the whole recursive map, including (5).

Continuous contraction now gives the rate by minimizing over its preimages. For a measurable map T and a reference probability ρ, the entropy contraction identity is

$$\inf_{\eta : T_\# \eta = \nu} D(\eta\|\rho) = D(\nu\|T_\#\rho). \tag{6}$$

The lower bound is data processing. When the right side is finite and $q = d\nu/d(T_\#\rho)$, the measure $d\eta = q(T(u))\, d\rho(u)$ attains equality. If the right side is infinite, data processing already proves the identity in the extended real sense. Apply (6) successively with $T = \mathsf{T}_{Q_0}$ and $T = \mathsf{T}_{K(\nu_{s-1})}$. The last minimization is

$$\inf_{v : Q^{1/2} v = z} \frac{1}{2}\|v\|_2^2 = \mathcal{G}_Q(z),$$

including singular Q. This proves (4).

The contraction theorem supplies a good rate and exponential tightness. For completeness, the displayed minimization is also directly well behaved: relative entropy is jointly lower semicontinuous, the map $\nu \mapsto \mathsf{M}_{K(\nu)}$ is weakly continuous, and

$$\mathcal{G}_Q(z) = \sup_{t \in \mathbb{R}^m} \left\{ t^\mathsf{T} z - \frac{1}{2} t^\mathsf{T} Q t \right\}$$

is jointly lower semicontinuous in (Q, z). Since $\mathcal{P}(E)^r$ is compact, every finite infimum in (4) is attained. Moreover $K(\nu) \preceq m I_m$, so $I(z) \geq \|z\|_2^2/(2m)$. □

Corollary A.2 (Full-margin masses and posterior rates)**.** *For a Boolean function f, regarded as a vector in $\{-1, 1\}^m$, set*

$$J(f) = \inf_{z :\ f_a z_a \geq 1/2\ \forall a} I(z).$$

Then the strict-margin event used throughout this paper has the exact rate

$$\lim_{N\to\infty} -\frac{1}{N}\log \mu_{N,1}(A_{N,f}) = J(f). \tag{7}$$

For a fixed noiseless training set S, let $\mathcal{B}_S$ be the Boolean functions agreeing with its labels. If $J_S = \min_{g\in\mathcal{B}_S} J(g) < \infty$, the actual globally conditioned posterior satisfies, for every $f \in \mathcal{B}_S$,

$$\lim_{N\to\infty} -\frac{1}{N}\log \Pi_{N,1}(f \mid S) = J(f) - J_S.$$

In particular, posterior mass concentrates on the minimizers of J within $\mathcal{B}_S$. These logarithmic asymptotics do not determine how mass is divided between tied minimizers.

Proof. For $c \geq 1$, retaining any feature laws in (4) and scaling only the output gives

$$I(cz) \leq c^2 I(z),$$

because the entropy terms are nonnegative and the Gaussian energy scales quadratically. If $f_a z_a \geq 1/2$ for all a, then cz has strict margin for every $c > 1$. Letting $c \downarrow 1$ shows that the infimum of I over the open strict-margin orthant equals its infimum over the closed orthant. The lower and upper large-deviation bounds therefore agree, proving (7).

Conditioning the prior on the global margin event and subsequently on the labels gives exactly

$$\Pi_{N,1}(f \mid S) = \frac{\mu_{N,1}(A_{N,f})}{\sum_{g\in\mathcal{B}_S} \mu_{N,1}(A_{N,g})}.$$

For fixed d this is a finite sum, whose exponential rate is the smallest $J(g)$ among its terms. This proves the ratio formula and concentration claim. □

References

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